



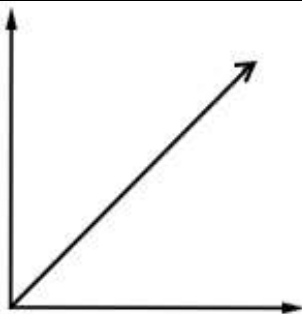
basic education

Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

**GEC PILOT STUDY
MARKING GUIDELINE 2023
MATHEMATICS PAPER1
GRADE 9**

SECTION A

- One mark per answer.
- There are no half marks.

No.		Expected answer		Clarification
1.	B	Irrational	✓	
2.	B	108	✓	$12 = 2 \times 2 \times 3$ $18 = 2 \times 3 \times 3$ $27 = 3 \times 3 \times 3$ $2 \times 2 \times 3 \times 3 \times 3 = 108$
3.	A	$\frac{2}{3}$	✓	Numerators $8 = 2^3$ $12 = 2^2 \times 3$ $\text{HCF} = 2 \times 2 = 4$ Denominators $30 = 2 \times 3 \times 5$ $36 = 2^2 \times 3^2$ $\text{HCF} = 2 \times 3 = 6$ $\text{HCF} = \frac{4}{6} = \frac{2}{3}$
4.	C		✓	

No.		Expected answer		Clarification
5.	A	103,13 km/h	✓	$s = \frac{d}{t}$ $120 = \frac{d}{2,75}$ $d = 120 \times 2,75$ $d = 330 \text{ km}$ $s = \frac{330}{3,2}$ $s = 103,13 \text{ km/h}$
6.	D	R1 339,42	✓	$A = P(1 + i)^n$ $A = R5\,265 \left(1 + \frac{12}{100}\right)^2$ $A = R5\,265(1 + 0,12)^2$ $A = R5\,265(1,12)^2$ $A = R5\,265(1,2544)$ $A = R6\,604,42$ $CI = R6\,604,42 - R5\,265$ $CI = R1\,339,42$
7.	C	$-7 \times (11 \times 4)$	✓	
8.	B	8	✓	$3(4) - 2(2) = 12 - 4$ $= 8$
9.	D	2 and $-\frac{1}{2}$	✓	
10.	C	14	✓	$\frac{2(-3) - (5) - 4(6 \div 8)}{5 - 6}$ $= \frac{-6 - 5 - 4\left(\frac{3}{4}\right)}{-1}$ $= \frac{-14}{-1}$ $= 14$
11.	A	1	✓	$\sqrt{49} - 2^3 + \sqrt[3]{216} \div 3$ $= 7 - 8 + 6 \div 3$ $= 7 - 8 + 2$ $= 1$
12.	C	$\frac{1}{5}$	✓	$\sqrt{\frac{\sqrt[3]{-64} + 5}{4^2 + 3^2}} = \sqrt{\frac{-4 + 5}{16 + 9}}$ $= \sqrt{\frac{1}{25}}$ $= \frac{1}{5}$
13.	A	a^5c	✓	$a^3 \times a^2c$ $= a^5c$

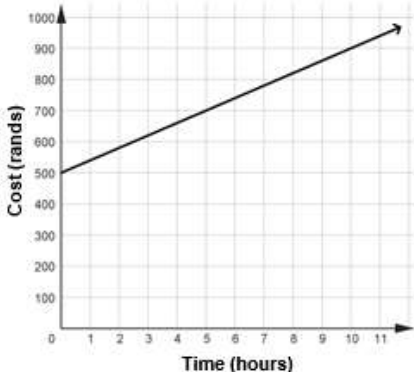
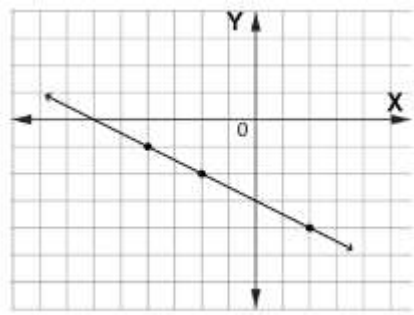
No.		Expected answer		Clarification
14.	C	2^{x+2}	✓	$\frac{4^{x+1}}{2^x}$ $= \frac{(2^2)^{x+1}}{2^x}$ $= \frac{2^{2x+2}}{2^x}$ $= 2^{2x+2-x}$ $= 2^{x+2}$
15.	B	$\frac{b+a}{ab}$	✓	$a^{-1} + b^{-1}$ $= \frac{1}{a} + \frac{1}{b}$ $= \frac{b+a}{ab}$
16.	B	1	✓	$(2^2)^3 \times (2^3)^{-2}$ $= 2^6 \times 2^{-6}$ $= 2^0$ $= 1$
17.	A	$-\frac{8x^3}{y^3}$	✓	$\left(\frac{x^{-3}y}{-2x^{-2}} \right)^{-3}$ $= \left(\frac{-2x^{-2}}{x^{-3}y} \right)^3$ $= \left(\frac{-2x^3}{x^2y} \right)^3$ $= \left(-\frac{2x}{y} \right)^3$ $= -\frac{8x^3}{y^3}$
18.	D	$\frac{7}{y}$	✓	$\frac{3(x+y)^0}{y} + 5y^{-1} - \frac{(x^{-1}y)^3}{x^{-3}y^4}$ $= \frac{3 \times 1}{y} + 5 \times \frac{1}{y} - \frac{x^{-3}y^3}{x^{-3}y^4}$ $= \frac{3}{y} + \frac{5}{y} - \frac{1}{y}$ $= \frac{7}{y}$
19.	B	35	✓	$3 + 5 ; 8 + 7 ; 15 + 9 ; 24 + 11 = 35$ OR Perfect squares – 1 $2^2 - 1 = 3$ $3^2 - 1 = 8$ $4^2 - 1 = 15$
20.	C	83	✓	$3 + 6 + 10 + 15 + 21 + 28 = 83$

No.		Expected answer		Clarification
21.	A	A constant ratio of three was used.	✓	
22.	C	$T_n = -2n + 7$	✓	1; 3; 5; 7; ... are not consecutive term positions 5; a; 1; b; -3; c; ... $a - 5 = 1 - a$ $2a = 6$ $\frac{2a}{2} = \frac{6}{2}$ $a = 3$ 5; 3; 1; ... are consecutive terms $3 - 5 = 1 - 3$ $d = -2$ $T_1 = -2(1) + __ = 5$ $T_2 = -2(2) + __ = 3$ $T_3 = -2(3) + __ = 1$ $T_n = -2n + 7$
23.	A	$3x^3$ and $3x(x^2)$	✓	
24.	D	-8 and 6^{th}	✓	$(-2x^2)^3$ $= (-2)^3(x^2)^3$ $= -8x^6$ Coefficient is -8 and degree is 6^{th}
25.	C	$x + 3x^2 + 2 - 4x$	✓	
26.	A	$-4x^3 - 2x^2 + x$	✓	$-x(4x^2 + 2x - 1)$ $= -4x^3 - 2x^2 + x$
27.	A	$3x^2 - 10x + 1$	✓	$3x(x - 4) + \frac{4x + 2}{2}$ $= 3x(x - 4) + \frac{4x}{2} + \frac{2}{2}$ $= 3x^2 - 12x + 2x + 1$ $= 3x^2 - 10x + 1$ OR $= \frac{2 \times 3x(x - 4) + 4x + 2}{2}$ $= \frac{6x^2 - 24x + 4x + 2}{2}$ $= \frac{6x^2}{2} - \frac{20x}{2} + \frac{2}{2}$ $= 3x^2 - 10x + 1$

No.		Expected answer		Clarification
28.	A	$\frac{x}{2}$	✓	$\sqrt[3]{0,125x^3}$ $= \sqrt[3]{\frac{1}{8}x^3}$ $= \sqrt[3]{\frac{1}{2^3}x^3}$ $= \left(\frac{1}{2^3}x^3\right)^{\frac{1}{3}}$ $= \left(\frac{1}{2^3}\right)^{\frac{1}{3}}(x^3)^{\frac{1}{3}}$ $= \frac{1}{2}x$ $= \frac{x}{2}$
29.	C	$2x^2 + \frac{17x}{3} - 1$	✓	$(x+3)(2x-\frac{1}{3})$ $= 2x^2 - \frac{x}{3} + 6x - \frac{3}{3}$ $= 2x^2 - \frac{x+18x}{3} - 1$ $= 2x^2 + \frac{17x}{3} - 1$
30.	D	16	✓	$-8\left(-\frac{1}{2}\right)^2 + 10(2) - 2$ $= -8\left(\frac{1}{4}\right) + 20 - 2$ $= -2 + 20 - 2$ $= 16$
31.	A	$(2x-3)(2x+3)$	✓	
32.	B	$(x-8)(x+3)$	✓	
33.	A	$6(y-3)(y+1)$	✓	
34.	B	$x+2$	✓	$\frac{2x^2-2x-12}{2x-6}$ $= \frac{2(x^2-x-6)}{2(x-3)}$ $= \frac{2(x+2)(x-3)}{2(x-3)}$ $= x+2$

No.		Expected answer		Clarification
35.	A	$x + y + 1$	✓	$\frac{(x + y)^2 - 1}{x + y - 1}$ $= \frac{(x + y - 1)(x + y + 1)}{x + y - 1}$ $= x + y + 1$
36.	A	2	✓	
37.	B	2	✓	$\frac{2p}{2} = \frac{4}{2}$ $p = 2$
38.	D	$x = 0$ or $x = 1$	✓	
39.	D	$b = 3$ or $b = -2$	✓	
40.	B	$2x - 8 = 18$	✓	
41.	C	4	✓	$y = 16 - 12$ $y = 4$
42.	C	-2	✓	$3^m = 9^{-1}$ $3^m = (3^2)^{-1}$ $3^m = 3^{-2}$ $m = -2$
43.	D	$p = 17$ and $t = 6$	✓	$y = x^2 + 1$ $p = (-4)^2 + 1$ $p = 16 + 1$ $p = 17$ and $y = x^2 + 1$ $37 = t^2 + 1$ $t^2 = 36$ $t = \pm 6$ $\therefore t = 6$
44.	D	$x = -6$ or $x = 1$	✓	$x^2 + 3x + 2x - 6 = 0$ $x^2 + 5x - 6 = 0$ $(x + 6)(x - 1) = 0$ $x = -6 \text{ or } x = 1$

No.		Expected answer		Clarification								
45.	B	$a = 0$ or $a = 4$	✓	$\left(\frac{1}{2}a - 1\right)\left(\frac{1}{2}a - 1\right) = 1$ $\frac{1}{4}a^2 - a + 1 = 1$ $\frac{1}{4}a^2 - a = 0$ $a^2 - 4a = 0$ $a(a - 4) = 0$ $a = 0$ or $a - 4 = 0$ $a = 0$ or $a = 4$ OR $\frac{1}{2}a - 1 = \pm 1$ $\frac{1}{2}a - 1 = 1$ or $\frac{1}{2}a - 1 = -1$ $\frac{1}{2}a = 2$ or $\frac{1}{2}a = 0$ $a = 4$ or $a = 0$								
46.	B	2 m	✓	$A = (x - 18)(x)$ $40 = x^2 - 18x$ $0 = x^2 - 18x - 40$ $0 = (x - 20)(x + 2)$ $x = 20$ or $x = -2$ $\therefore x = 20$ breadth = $20 - 18$ breadth = 2 m								
47.	A	$b = 33$	✓	$y = 4x - 3$ $b = 4(9) - 3$ $b = 33$								
48.	C	<table border="1"><tr><td>x</td><td>-2</td><td>0</td><td>2</td></tr><tr><td>y</td><td>-8</td><td>-2</td><td>4</td></tr></table>	x	-2	0	2	y	-8	-2	4	✓	
x	-2	0	2									
y	-8	-2	4									
49.	B	$m = (n + 1)^2$	✓	$m = (\text{Even} + 1)^2$ $m = (\text{Odd})^2$ $m = \text{Odd}$								
50.	D	$y = -\frac{1}{2}x + 1$	✓	$m = -\frac{1}{2}$ y-intercept at (0; 1) and $c = 1$ $y = -\frac{1}{2}x + 1$								
51.	D	$V = 4t + 6$	✓	Common difference = 4 $V = 4t + c$ By inspection: $V = 4t + 6$								

No.		Expected answer	Clarification										
52.	A		✓ $c = 40t + 500$ y -intercept = R500 and the gradient = 40										
53.	B	$(0; -5)$	✓										
54.	D	1	✓ $y = 2x - 2$ $2x - 2 = 0$ $2x = 2$ $x = 1$										
55.	C	$-\frac{3}{4}$	✓										
56.	B		✓ <table border="1" data-bbox="888 1034 1457 1151"><tr><td>x</td><td>-4</td><td>-2</td><td>2</td></tr><tr><td>y</td><td>-1</td><td>-2</td><td>-4</td></tr></table>	x	-4	-2	2	y	-1	-2	-4		
x	-4	-2	2										
y	-1	-2	-4										
57.	C	<table border="1" data-bbox="338 1348 753 1464"><tr><td>x</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>y</td><td>12</td><td>20</td><td>28</td><td>36</td></tr></table>	x	1	2	3	4	y	12	20	28	36	✓ Constant difference of 8
x	1	2	3	4									
y	12	20	28	36									
58.	D	$-2x + y - 2 = 0$	✓ $-2x + y - 2 = 0$ $y = 2x + 2$										
59.	B	$(-1; -3)$	✓ $y = 2x - 1$ $-3 = 2(-1) - 1$ $-3 = -3$										
60.	A	$y = -3x + 5$	✓ $m_{AB} = m_{CD}$...Parallel lines $y = -3x + c$ $\therefore 2 = -3(1) + c$ $c = 5$										
			Section A total										
			60										

SECTION B

- Do not penalise the learner for the same mistake twice.
- *There are no half marks.*

MARKING CODES	
M	Method mark
A	Accuracy mark
CA	Consistent Accuracy mark

No.	Expected answer	Clarification	Mark
61.	$y = 3x - 5 \checkmark M$ $46 = 3q - 5 \checkmark M$ $q = \frac{51}{3}$ $q = 17 \checkmark CA$	General term: 1 mark Substitution: 1 mark Answer: 1 mark Accept $x = 17$ Answer only full marks	3
62.	$\frac{-5y^3(6y-1) - 3y^2(5-10y^2)}{5y^2}$ $= \frac{-30y^4 + 5y^3 - 15y^2 + 30y^4}{5y^2} \checkmark M$ $= \frac{5y^3 - 15y^2}{5y^2}$ $= \frac{5y^2(y-3)}{5y^2} \checkmark M$ $= y - 3 \checkmark CA$ OR $\frac{y^2(6y-1)}{-y} - \frac{3y^2(5-10y^2)}{5y^2}$ $= \frac{6y^3 - y^2}{-y} - \frac{15y^2 - 30y^4}{5y^2} \checkmark M$ $= \frac{6y^3}{-y} - \frac{y^2}{-y} - \frac{15y^2}{5y^2} + \frac{30y^4}{5y^2}$ $= -6y^2 + y - 3 + 6y^2 \checkmark M$ $= y - 3 \checkmark CA$	Simplification: 1 mark $\frac{5y^2(y-3)}{5y^2}$: 1 mark Answer: 1 mark Simplification: 1 mark $-6y^2 + y - 3 + 6y^2$: 1 mark Answer: 1 mark	3

No.	Expected answer	Clarification	Mark
63.	Let x be the mass of the car $900 + 120 + \frac{x}{5} = x \checkmark \mathbf{M}$ $4500 + 600 + x = 5x \checkmark \mathbf{M}$ $4x = 5100$ $x = 1275 \text{ kg } \checkmark \mathbf{CA}$ $\therefore \text{The car's engine} = \frac{x}{5}$ $= 255 \text{ kg } \checkmark \mathbf{CA}$ OR Let x be the mass of the car $900 + 120 + \frac{x}{5} = x \checkmark \mathbf{M}$ $900 + 120 = \frac{4}{5}x \checkmark \mathbf{M}$ $4x = 5100$ $x = 1275 \text{ kg } \checkmark \mathbf{CA}$ $\therefore \text{The car's engine} = \frac{x}{5}$ $= 255 \text{ kg } \checkmark \mathbf{CA}$	Equation: 1 mark Simplification: 1 mark $x = 1275 \text{ kg}$: 1 mark Answer: 1 mark Equation: 1 mark Simplification: 1 mark $x = 1275 \text{ kg}$: 1 mark Answer: 1 mark	4
64.			
64.1	On graph $\checkmark \checkmark \checkmark \mathbf{A}$	Correct plotting of any 2 points: 1 mark Correct y-intercept: 1 mark Shape of graph: 1 mark	(3)
64.2	On graph $\checkmark \mathbf{A}$	Correct drawing of graph: 1 mark	(1)

No.	Expected answer	Clarification	Mark
64.3	Distance between x - intercepts = 6 units ✓CA	Answer: 1 mark	(1)
Section B total			15